



Ransomware: Trends, Impact, and Defensive Strategies

CSF3233 Cyber Security
Report
Prepared for: Dr. Firdaus Afifi bin Md Isa

Group 5 Members:

Matric No.	Name
S70012	Muhammad Syakil bin Mohd Zaidi
S70507	Khairul Azwan Bin Mohamad Ali
S71282	Muhamad Alif Bin Abd Satar

NotebookLM Link:

<https://notebooklm.google.com/notebook/8fa99f47-e646-412b-8722-139ddf411b85>

Table of contents

1 Introduction.....	2
1.1 Evolution of Ransomware Threats.....	2
2 Current Trends in Ransomware.....	3
2.1 Industrialisation of Ransomware-as-a-Service (RaaS).....	3
3 Comprehensive Impact Analysis.....	5
3.1 Financial and Economic Impact.....	5
3.2 Operational and Systemic Disruption.....	5
3.3 Sector-Specific Impact Analysis.....	6
3.4 Artificial Intelligence and Integrity Ransomware.....	6
3.5 Reputational and Legal Impact.....	7
4 Holistic Defensive Strategies.....	8
4.1 Multi-Layered Resilient Defence Models.....	8
4.2 Artificial Intelligence and Machine Learning for Detection.....	8
4.3 Protecting Artificial Intelligence Assets from Integrity Attacks.....	9
4.4 Defensive Strategies for Small and Medium Enterprises.....	9
4.5 Regulatory Compliance and Incident Reporting.....	10
5 Synthesis and Conclusion.....	11

1 Introduction

1.1 Evolution of Ransomware Threats

Ransomware has transitioned from a localized, opportunistic form of malicious software into a highly organized, transnational digital extortion ecosystem that poses an immediate threat to modern enterprise architectures, cloud environments, and Critical National Infrastructure (CNI). Initially emerging in the late 1980s as basic file-locking mechanisms, contemporary ransomware strains have evolved into sophisticated payloads that leverage complex deployment pipelines, stealth encryption protocols, and multi-vector coercion strategies. The rapid institutional adoption of digital software frameworks, decentralized cloud networks, and automated data pipelines has exponentially expanded the organizational attack surface, allowing malicious actors to target high-value assets with remarkable precision. This evolution is heavily accelerated by macro-economic cybercrime syndicates utilizing the Ransomware-as-a-Service (RaaS) business model, which dramatically lowers the technical barrier to entry for threat actors while enabling the deployment of polymorphic code capable of evading traditional, signature-based security controls.

The primary challenge in managing modern ransomware vectors lies in the structural fragility of conventional asset-centric protection assumptions. Historically, security frameworks have operated under a data- and code-centric paradigm, presupposing that system safety can be maintained strictly through file-level integrity checks, static rule-based firewalls, and isolated database backups. However, advanced threat dynamics such as intermittent file encryption, which selectively corrupts minute data blocks to evade baseline entropy tracking, and automated process hollowing actively circumvent these legacy mechanisms. Furthermore, within complex environments integrating artificial intelligence and automated decision-making engines, a restored system can pass initial file-level validation checks while remaining behaviorally compromised due to deep-seated payload anomalies or dataset corruption. Consequently, there is a critical need to transition national cyber-defence strategies away from reactive, post-incident data restoration toward proactive, multi-layered resilience models that prioritize continuous behavioural verification and rapid containment protocols.

This report presents a comprehensive academic analysis of the contemporary ransomware landscape, specifically aligning empirical data with innovative defensive frameworks designed to safeguard volatile digital environments. By examining the core technical mechanisms driving modern extortion cycles, the subsequent sections evaluate the changing operational trends of threat actors, including double and triple extortion tactics that exploit supply chain dependencies and mandatory compliance reporting regimes. Additionally, the paper investigates the broader financial, systemic, and reputational impacts of these disruptions across both small-and-medium enterprises (SMEs) and large-scale public institutions. Ultimately, this study synthesizes cutting-edge defensive interventions ranging from deep-learning sequence sensors and reinforcement learning blocking agents to behavior-aware integrity protection systems establishing a forward-compatible blueprint for holistic cyber resilience.

2 Current Trends in Ransomware

2.1 Industrialisation of Ransomware-as-a-Service (RaaS)

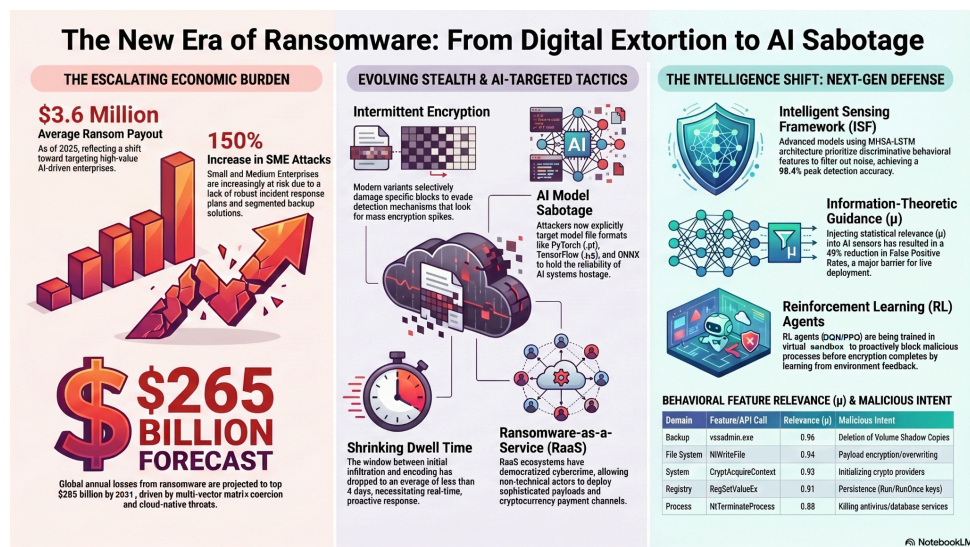


Figure 1 Ransomware Trend Infographic

The contemporary cyber-threat landscape is characterised by a profound structural shift in the operational models, execution velocities, and strategic objectives of ransomware syndicates. Central to this evolution is the industrialisation of the Ransomware-as-a-Service (RaaS) business model, a decentralized ecosystem that democratises cybercrime by allowing non-technical affiliates to lease sophisticated, pre-configured payloads while utilizing automated cryptocurrency channels to execute financial extortion. This institutionalised division of labour has led to an aggressive expansion in target selection, causing a documented 150% increase in attacks directed at small and medium-sized enterprises (SMEs). Because smaller organisations frequently operate under acute resource constraints, they often exhibit severe defensive vulnerabilities, such as unsegmented backup architectures and a systemic absence of formalized incident response plans, making them highly lucrative targets for RaaS affiliates. This macro-economic trend has driven global annual losses toward a forecast exceeding \$265 billion by 2031, as visualised in the conceptual framework of Figure 1.

Concurrently, the technical execution of ransomware payloads has achieved a level of stealth that actively invalidates legacy host-based monitoring configurations. A primary indicator of this sophistication is the critical minimization of attacker dwell time, the temporal window between initial network infiltration and payload deployment which has dropped to an operational average of less than four days. Within this compressed timeline, modern variants increasingly deploy intermittent encryption algorithms. Rather than encrypting a target file in its entirety, which triggers immediate security alerts by generating massive data entropy spikes, intermittent encryption selectively corrupts or scrambles specific data blocks at randomized offsets. This technique preserves the overarching file structure and header metadata, effectively tricking traditional signatureless detection systems and regularized scanners into validating the malicious payload as a benign background process while permanently destroying the underlying data integrity.

Beyond the disruption of standard corporate files, contemporary extortion strategies have expanded into specialized algorithmic sabotage, specifically targeting the operational infrastructure of artificial intelligence (AI) and machine learning systems. As demonstrated in recent forensic analyses, advanced threat actors now deliberately scan environments to identify, exfiltrate, and encrypt proprietary AI model formats, explicitly targeting PyTorch (.pt), TensorFlow (.h5), and ONNX (.onnx) frameworks to hold an organisation's core decision-making logic hostage. This manifestation of "integrity ransomware" moves beyond simple data availability denial; by subtly poisoning training pipelines or injecting conditional backdoors into model checkpoints, attackers can cause an AI system to fail or execute malicious instructions only when exposed to highly specific, trigger-based user inputs. Because the compromised model structurally maintains normal accuracy levels during baseline testing, the distortion remains invisible to standard security gates, forcing public and private institutions to confront high-leverage extortion cycles where the reliability of autonomous assets is held completely captive.

3 Comprehensive Impact Analysis

3.1 Financial and Economic Impact

The financial implications of ransomware have increased far beyond the usual payment of ransom demands, it evolves into a macro-economic threat that organizations, industries and national economies have been affected. The complexity of ransomware campaigns through the escalation of the Ransomware-as-a-Service (RaaS) model contribute to a rise in global cybercrime-related losses. Industry forecasts estimate that ransomware and associated cybercrime activities could generate annual global damages exceeding USD 265 billion by 2031. This escalation is mostly driven by the growing frequency of attacks, increasing ransom demands and the significant costs associated with incident recovery and business continuity restoration.

These days ransomware attacks enforce multiple layers of financial burden upon victim organizations. Although ransom payments are still a big problem, the costs of getting back into the system often go beyond the ransom itself. Costs for digital forensics, system restoration, legal advice, regulatory compliance, cyber insurance claims, and business interruption can make the overall financial effect much bigger. Recent studies show that the average cost of recovery after a ransomware attack is more than USD 1.5 million, even if the company doesn't pay the fee. Nowadays cybercriminals are going after more and more valuable digital assets, such as private AI models and machine learning datasets. Rebuilding and teaching these assets can cost hundreds of millions of dollars. Because of this, ransomware has become a strategic economic weapon that can stop businesses from growing, make investors lose faith, and hurt their ability to stay in business in the long run.

3.2 Operational and Systemic Disruption

In addition to financial impact, ransomware attacks can cause significant operational issues, potentially impacting the continuity of critical business systems and operations. File encryption is not the only thing ransomware can inflict on an organizational ecosystem, because there are new variants in development that are designed to wreak havoc on networks, cloud applications and operational technologies, which are automated functions tasked with controlling physical equipment. With the dwindling of the attackers' dwell time to under four days, addressing the issue becomes a challenge for the cyber security team(s) to correct the issue before harm can be done.

For Critical National Infrastructure (CNI) related industry, especially energy production, transportation, healthcare, telecommunications and water management systems, the impacts on operations are particularly severe. If these sectors are successfully attacked, they can create service outages, disruptions to supply and jeopardize public safety. Furthermore, often times, today's ransomware attacks are leveraging supply chain vulnerabilities in a "triple extortion" attack. By which I mean strike a particular service/software vendor and then spread out to affect thousands of producers or services that rely on that victim's services. The domino effect highlights that ransomware is from a simple cyber threat to a systemic risk that can have serious economic impacts on a region or a nation.

3.3 Sector-Specific Impact Analysis

Ransomware affects different industries in different ways, and they vary based on how important data is to the business, how well prepared they are to defend against attacks and whether or not the industry has specific requirements for its operations. Unsurprisingly, the healthcare industry is among the top targets for cyberattacks, given its reliance on the timely access to patient data and systems that could prove to be life-or-death. An interruption in healthcare can lead to a delay in medical care, disruption to emergency services and potentially put lives at risk. The healthcare industry continues to be a target for ransomware, with reports of ransomware being the primary cause of most significant healthcare industry cyber incidents in multiple countries.

Ransomware operators are also increasingly targeting Small and Medium Enterprises (SMEs) as appealing targets. The growth of this RaaS has largely led to larger campaigns against smaller companies that may not have their own cybersecurity staff, monitoring tools, or incident response capabilities. This can have a disproportionate impact on SMEs, resulting in extended downtime, fewer customers and in many instances, the closure of the business.

The financial industry is also a critical sector because of the high volume of assets that are managed on it and the confidential and vital information that the customers provide. For financial institutions, ransomware attacks may impact their online banking systems, payment rails and internal functions, thus impacting their economic bottom line as well as customer risks. This may taint investments in digital financial infrastructures and create systemic financial risk if such disruptions occur.

3.4 Artificial Intelligence and Integrity Ransomware

One of the more worrying trends in the ransomware market is the increase of integrity ransomware that is not just focussing on access to data, but also on the reliability, or trustworthiness, of AI systems. While traditional ransomware attacks aim to encrypt files and disrupt services, integrity ransomware attacks are designed to deceive, “poison” and “corrupt” decision-making processes in AI and machine learning algorithms.

Popular formats like PyTorch (.pt), TensorFlow (.h5), and ONNX (.onnx) files are common targets for threat actors as they can be a major intellectual and financial investment. AI models can undergo subtle changes in parameters or be susceptible to certain triggers that result in erroneous outputs, but work well in normal validation tests. AI systems can be engineered to produce false results under specific conditions, leaving them to execute normally during routine validation testing. This allows it to slip through the cracks in detection.

The impact of integrity attacks is vast. It is known that small changes to the parameters in AI models can lead to significantly degrading results, with the models

providing unreliable predictions, recommendations, or actions as a result of corruption. This is a new form of “extortion” that could be the trust conveyed in the technology itself, especially for organizations that depend on AI-driven decision-making functions in fields of medicine, finance, manufacturing, or autonomous systems. In this regard, integrity ransomware poses a significant threat to the future of AI technologies, which calls for effective behavioral verification in addition to traditional cybersecurity methods.

3.5 Reputational and Legal Impact

The long-term consequences of ransomware incidents frequently extend beyond immediate technical and financial recovery. One of the most significant outcomes is reputational damage, which can erode stakeholder confidence and diminish public trust in an affected organization. Customers, business partners, and investors may perceive a successful ransomware attack as evidence of inadequate cybersecurity governance, resulting in reduced market competitiveness and long-term brand degradation.

In addition to reputational harm, organizations increasingly face substantial legal and regulatory obligations following a ransomware incident. Contemporary regulations such as the European Union's Network and Information Security Directive (NIS2) and other national cybersecurity frameworks require organizations to implement appropriate security controls and promptly report significant cyber incidents. Failure to comply with these requirements may result in substantial financial penalties, regulatory investigations, and legal liabilities.

Furthermore, modern double-extortion tactics significantly increase legal and reputational risks by combining data encryption with data exfiltration. Threat actors often threaten to publicly release sensitive customer information, proprietary business data, or intellectual property if ransom demands are not met. Such disclosures may trigger privacy violations, contractual disputes, and intellectual property losses that persist long after technical recovery has been completed. As a result, ransomware incidents have become a multidimensional organizational crisis involving technological, financial, legal, and reputational consequences that require comprehensive risk management strategies.

4 Holistic Defensive Strategies

4.1 Multi-Layered Resilient Defence Models

Conventional security architectures based on signatures and backup independent of the system have proven ineffective in the face of today's ransomware variants that use intermittent encryption and process hollowing. To achieve this, a stronger defence strategy is the adoption of a multi-layered resilient defence model, which combines preventive, detective, and responsive controls throughout the attack lifecycle. In Matnale (2026) a holistic approach is described, including pre-emptive threat hunting with real-time behavioural monitoring, allowing organisations to uncover and thwart malicious activity before an encryption payload is dropped.

A key piece to this would be network segmentation, preventing data from being laterally moved; immutable storage for backups, where data cannot be encrypted or removed; and continuous checks on data integrity using entropy-based anomaly detection. The framework also highlights the need for automated incident response playbooks which can be triggered as soon as an attack is detected, thereby decreasing the "dwell time" that the threat actors currently have. Managed security service providers (MSSPs) can offer these capabilities to smaller organisations with fewer security resources in a subscription-based manner, thereby making enterprise-class security accessible for everyone.

4.2 Artificial Intelligence and Machine Learning for Detection

AI-based defense methods have the potential to detect ransomware attacks early, even against new and polymorphic variants that are not easily identified by traditional signatures, offering immense promise. In an intelligent sensing framework introduced by Alqahtani (2025), the use of Multi-Head Self-Attention Long Short-Term Memory (MHSA-LSTM) machine learning architectures is proposed to analyse system behaviour patterns and detect anomalies related to ransomware with high accuracy. These deep-learning models are highly effective at identifying slight variations in file access and encryption activity, and network traffic that precede ransomware attacks.

Moreover, reinforcement learning agents are a new area in ransomware proactive defense. In this case, Thakur (2025) shows the possibility of training these agents to dynamically detect ransomware without the need for pre-known specific signatures. Reinforcement learning models are more flexible than static rule-based models, which evolve over time as they learn from detecting successful and missed attacks. This adaptive capability is especially beneficial for attacks that involve merely some blocks being encrypted, a tactic that is sometimes used, so it helps to bypass the typical entropy-based attack detection method. But these AI-powered systems come with a high cost of computation, and organizations need to be careful not to impact system performance in their production environments.

4.3 Protecting Artificial Intelligence Assets from Integrity Attacks

The development of integrity ransomware targeting AI models is a major security vulnerability in existing defense mechanisms, as pointed out in the impact assessment. Under the new age of AI, Min (2025) underscores the need for special protection measures specifically designed for machine learning assets, such as model encryption, cryptographic signing to confirm the integrity of the model, and regular behavioural validation tests to identify small parameter adjustments. It is crucial that organisations that are building or implementing an AI system treat the model files in the same way as they treat their source code or sensitive customer data, meaning that they have to have strict access restrictions and audit trails.

To detect that AI systems have been compromised and able to pass standard validation tests despite the presence of conditional backdoors or poisoned training data, behavioural verification protocols are necessary. The protocols also include testing AI models with a variety of input scenarios, such as edge cases and adversarial examples, to ensure that the models make the same decisions in different conditions. Moreover, organisations need to store several copies of important AI models to quickly restore them to a known good status when suspected integrity breaches occur. This fits into the concept of zero trust, which means that no part of a system is inherently trusted and ongoing verification is required.

4.4 Defensive Strategies for Small and Medium Enterprises

SMEs are especially vulnerable to ransomware attacks because they may have tight budgets, lack security expertise, and poorly designed backup systems. Awan (2025) systematically maps cybersecurity threats and defensive strategies particularly for small and medium enterprises (SMEs), and makes a list of interventions that are cost-effective and provide high levels of protection for every dollar invested. These include essential security habits like regular software patching, multi-factor authentication and employee security awareness training, all of which cover the most frequent entry points for ransomware attacks.

The need for immutable backup solutions is especially important for SMEs. Cloud-based backup services that support versioning and retain backups from ransomware even if they are deleted or encrypted provide cost-effective protection against ransomware data loss. In addition, SMEs should create and continually test incident response plans suitable to their operations, and make sure that key employees know their roles and responsibilities if they become victims of ransomware attacks. SMEs can also benefit from timely, up-to-date threat intelligence sharing through industry associations or regional business networks without the need for dedicated research skills.

4.5 Regulatory Compliance and Incident Reporting

Regulatory requirements for ransom attacks are becoming tougher, and the UK NIS Regulations and the EU NIS2 Directive require organisations affected by ransom attacks to comply with reporting requirements. Ali (2026) explores the effectiveness of these regulatory regimes and highlights how compliance requirements can foster accountability and promote security enhancements, but also present legal and reputational threats that need to be proactively addressed. Organisations should set up clear incident reporting procedures that strike the right balance between regulatory requirements and the need to investigate and contain attacks prior to public disclosure.

Good defense strategies include legal preparation as a part of the process, and proper handling of evidence, complete communications with regulators, and preservation of legal privileges as necessary. In addition, emerging regulations that focus specifically on AI cybersecurity will impose further obligations on organisations using machine learning systems, such as having to document security controls and regularly independent audit of AI assets, as suggested by Ramli (2026). When building a defensive architecture, it's important for organisations to incorporate regulatory compliance, which decreases the risk of the successful attack and the impact of an attack when one does take place.

5 Synthesis and Conclusion

5.1 Synthesis of Key Findings

This report has identified the transition of the threat of ransomware from a relatively primitive file encryption threat, into a sophisticated and highly organized cybercrime ecosystem. The development of the Ransomware-as-a-Service (RaaS) model has made possible the rapid scaling of ransomware attacks, as threat actors have a lower technical threshold to initiate ransomware operations. Concurrently, the increase in potency of increasing attack methodologies such as intermittent encryption, reduction of attacker dwell time and 2- and 3- times extortion, have made ransomware campaigns considerably more effective.

Furthermore, upon analysing, we learn that the threat of ransomware goes way beyond only denial of access to files. Currently, ransomware attack pose a threat to entire organizational environments; from cloud environments to supply chains, and vital national infrastructure. Most recently we see on the emergence of integrity ransomware, which showcases an increased ability of ransomware firms to attack not data, but the fabric of artificial intelligence and it relies upon.

5.2 Relationship Between Impact and Defensive Strategies

The threats exposed by this report are a reminder to organizations that holistic, adaptive defenses are paramount. The threat is both financial as well as operational and sector specific. And while brand, regulatory and legal fallout from a ransomware attack are also significant issues, it's important to note that this is a business level, organizational level issue, not an IT level issue alone. In today's threat landscape signature based-attack protection and "back-and-Restore" type data recovery strategies are not adequate, and must evolve as the ransomware threat continues to mature.

Companies need to invest in the layered "resilient" defenses including: Prevention, Detection, Response, and Recovery. Multi-layered defense in depth strategy, behavioral analytics and the intelligent-driven capabilities in today's protection platforms and AI based-attack identification systems will offer organizations better defenses. Those that incorporate AI into their solutions must focus on the protection of AI models as well as data datasets from integrity attacks. And employees, regulations, and robust incident response plans are an integral part of the solution.

5.3 Conclusion

Ransomware remains a leading global cybersecurity threat. The growth of Ransomware-as-a-Service (RaaS), the use of ever more cunning and complex techniques, and the development of new, AI-powered extortion schemes have repositioned the threat of ransomware far beyond IT departments and into the realm of economic vitality, public confidence, and national security. What the findings of this report highlight is the multi-faceted impact of ransomware; it damages financial performance, operational resilience, brand reputation, and regulatory status.

Furthermore, the greater organizations rely on cloud, IT infrastructure, and artificial intelligence, the more the risk of ransomware events are amplified. It is clear then, that the future strategy for cybersecurity needs to shift from incident response to proactively built resilience and defense. The integration of advanced detection tools, real-time monitoring, secure data backups, AI asset protection, and comprehensive governance procedures, can considerably mitigate a ransomware-driven incident. Building cybersecurity resilience, therefore, is vital to ensuring operational resilience for organizations, going forward, in our highly interconnected world.